

Longer Contexts Reduce Factual Reliability

Separating Hallucinatory and Grounded Errors in Llama 3.2 3B

Experiment D Final Report

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Model: meta-llama/Llama-3.2-3B-Instruct

Benchmark: 500 question families, six context lengths

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Executive Summary

This study asks how increasing long-context length affects factual reliability when an LLM must answer factual questions from authentic but competing primary-source records. The benchmark used 500 question families across SEC, FDA/Drugs@FDA, ClinicalTrials.gov, and FRED/ALFRED. Each family was evaluated at 4K, 8K, 16K, 32K, 64K, and an empirically hardware-bounded 82K condition.

Overall inaccuracy increased from 49.6% at 4K to 70.7% at 82K. GEE OR per 2x rendered context = 1.232, 95% CI [1.178, 1.288], $p = 8.66 \times 10^{-20}$. Grounded Inaccuracy increased strongly. Hallucinatory Inaccuracy increased in the full dataset, but after excluding UNANSWERABLE tasks it decreased, indicating the full-dataset hallucination trend is driven primarily by failed abstention.

1. Introduction and Motivation

The central research question is how increasing context length affects factual reliability when an LLM must retrieve and reason over authentic but competing contextual records. A single hallucination rate is insufficient because long-context failures may come from unsupported fabrication, selection of the wrong legitimate contextual fact, wrong entity/temporal/version binding, calculation mistakes, or failure to abstain.

Every successful response is classified as Correct, Hallucinatory Inaccuracy, or Grounded Inaccuracy. Inaccuracy is the sum of Hallucinatory Inaccuracy and Grounded Inaccuracy. Runtime failures are separate and are not factual inaccuracies.

2. Data Sources

The benchmark uses four authoritative primary-source domains. Authentic records were used to construct target evidence, deterministic gold answers, same-domain distractors, temporal competitors, version competitors, entity competitors, series/unit competitors, and unanswerable cases.

Table 1. Authoritative source domains.

Domain	Families	Role
SEC	125	Company financial filings,

		concepts, periods, units, versions
FDA / Drugs@FDA	125	Drug application/product records, strengths, dosage forms, routes
ClinicalTrials.gov	125	Trial identifiers, statuses, dates, arms, posted results
FRED / ALFRED	125	Economic time series, vintages, units, seasonal/series variants

3. Question-Family Design

The benchmark contains 400 answerable families and 100 unanswerable families. The same question, gold answer, evidence policy, and answerability are preserved across all context lengths within a family.

Table 2. Question-type distribution.

Question Type	Families	Purpose
DIRECT_RETRIEVAL	100	Retrieve one explicitly requested fact from competing records
RETRIEVAL_CALCULATION	150	Retrieve multiple operands and compute a deterministic result
TEMPORAL_VERSION	55	Select the correct date, period, or version among competitors
ENTITY_UNIT_BINDING	95	Bind a requested value to the correct entity, unit, or series
UNANSWERABLE	100	Return INSUFFICIENT_EVIDENCE when required evidence is absent
Total	500	

4. Context-Length Design

The context ladder is 4K, 8K, 16K, 32K, 64K, and 82K. The 82K condition is not a doubling of 64K; models therefore used actual log2 rendered input tokens.

Table 3. Mean rendered input tokens by context condition.

Context	Mean Rendered Input Tokens
4K	4,273.3
8K	8,330.6
16K	16,442.3
32K	32,671.4
64K	65,126.1
82K	81,745.1

5. Experiment Size and Runtime Outcome

The frozen benchmark has 3,000 attempted inference instances. There were 2,998 successful generations and two CUDA OOM runtime failures, both at 82K. The OOM cases were excluded from factual-outcome denominators.

6. Model and Inference Configuration

The model was meta-llama/Llama-3.2-3B-Instruct, revision 0cb88a4f764b7a12671c53f0838cd831a0843b95, run on an NVIDIA GeForce RTX 4090 with BF16, batch size 1, standard Hugging Face DynamicCache, greedy decoding,

do_sample=False, num_beams=1, max_new_tokens=128, no quantization, no cache offloading, and no model offloading. The prompt was llama_chat_v4, prompt hash 5d2869822989e19b, with frozen date 09 Aug 2026. Successful outputs had zero malformed outputs, zero 128-token cap hits, and zero repetitive/degenerate outputs.

7. Grading

Grading was deterministic. The frozen grader hash was d9282a0ccc50daba3bfd232c058dfbab63a5c19a7323d585a7c2236b3a6c4ba8. Nineteen cases were manually adjudicated under the frozen rules: seven correct grader edge cases and twelve grounded WRONG_ENTITY cases. No manual case was converted to hallucination.

Table 4. Final factual-outcome totals.

Outcome	Count
Correct	1176
Inaccurate	1822
Hallucinatory Inaccuracy	1132
Grounded Inaccuracy	690
Runtime Failures	2
Ambiguous	0

8. Primary Factual-Reliability Results

Table 5. Primary factual-reliability results by context.

Context	Gradable N	Correct	Inaccurate	Inaccuracy 95% CI	Hallucinatory Inaccuracy	Grounded Inaccuracy	Runtime Failures	Mean Latency
4K	500	50.4%	49.6%	45.2%-54.0%	34.2%	15.4%	0	0.316 s
8K	500	45.6%	54.4%	50.0%-58.8%	37.2%	17.2%	0	0.598 s
16K	500	41.2%	58.8%	54.4%-63.2%	34.8%	24.0%	0	1.212 s
32K	500	38.4%	61.6%	57.4%-65.8%	38.2%	23.4%	0	2.842 s
64K	500	30.4%	69.6%	65.6%-73.6%	41.8%	27.8%	0	7.827 s
82K	498	29.3%	70.7%	66.6%-74.7%	40.4%	30.3%	2	11.260 s

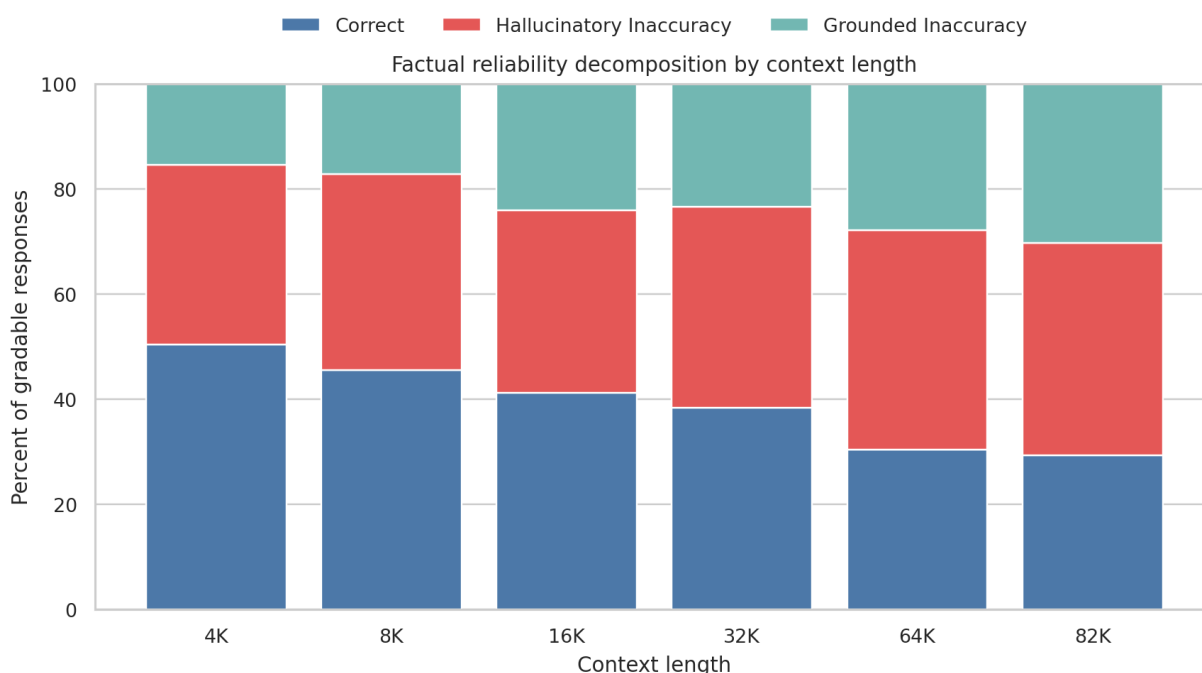


Figure 1. Factual Reliability Decomposition. Correct, Hallucinatory Inaccuracy, and Grounded Inaccuracy sum to 100% of gradable responses at each context length.

9. Primary Statistical Model

The primary model was GEE logistic regression with outcome inaccurate, predictor $\log_2(\text{rendered_input_tokens})$, and clustering by question_family_id. Each +1 increase corresponds to a doubling of rendered context. OR = 1.232, 95% CI [1.178, 1.288], $p = 8.66 \times 10^{-20}$. This is an odds increase, not a percentage-point increase.

Table 6. GEE trend models.

Outcome	OR per 2x Context	95% CI	p-value	Interpretation
Overall Inaccuracy	1.232	[1.178, 1.288]	8.66e-20	Odds ratio per doubling of rendered input tokens
Hallucinatory Inaccuracy	1.070	[1.027, 1.115]	0.00133	Odds ratio per doubling of rendered input tokens
Grounded Inaccuracy	1.215	[1.153, 1.280]	3.34e-13	Odds ratio per doubling of rendered input tokens
Hallucinatory vs Grounded Among Inaccuracies	0.936	[0.897, 0.977]	0.00249	Odds ratio per doubling of rendered input tokens

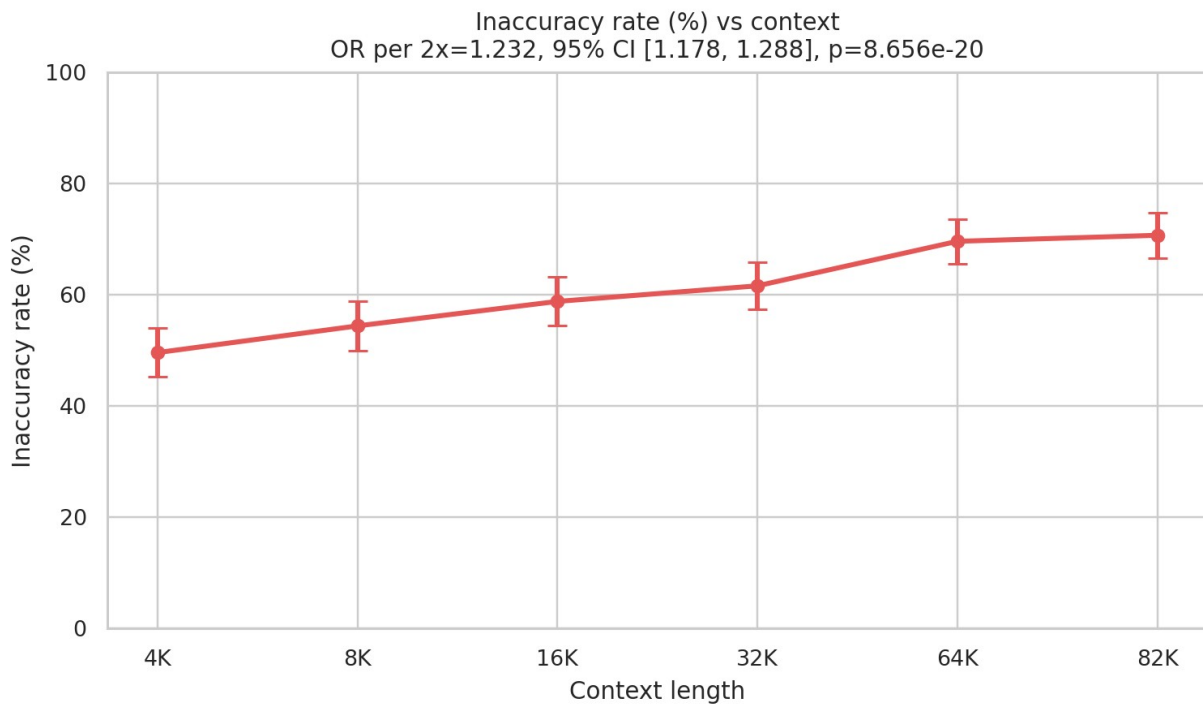


Figure 2. Overall Inaccuracy vs Context. GEE OR = 1.232 per 2x rendered context, 95% CI [1.178, 1.288], $p = 8.66 \times 10^{-20}$.

10. Hallucinatory Inaccuracy

Hallucinatory Inaccuracy increased in the full benchmark from 34.2% at 4K to 40.4% at 82K. GEE OR = 1.070, 95% CI [1.027, 1.115], $p = 0.00133$. This full-dataset result must be interpreted with the UNANSWERABLE sensitivity analysis.

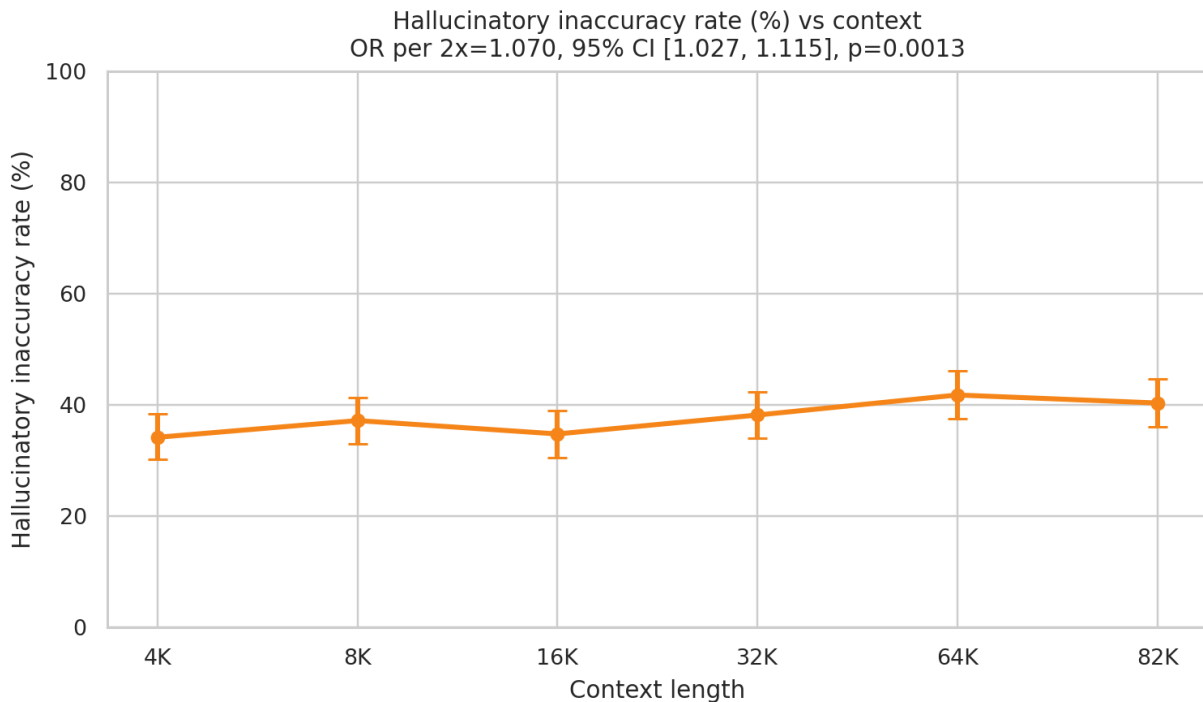


Figure 3. Hallucinatory Inaccuracy vs Context. GEE OR = 1.070, 95% CI [1.027, 1.115], $p = 0.00133$.

11. Grounded Inaccuracy

Grounded Inaccuracy increased from 15.4% at 4K to 30.3% at 82K. GEE OR = 1.215, 95% CI [1.153, 1.280], $p = 3.34 \times 10^{-13}$. These are legitimate contextual values bound or reasoned over incorrectly.

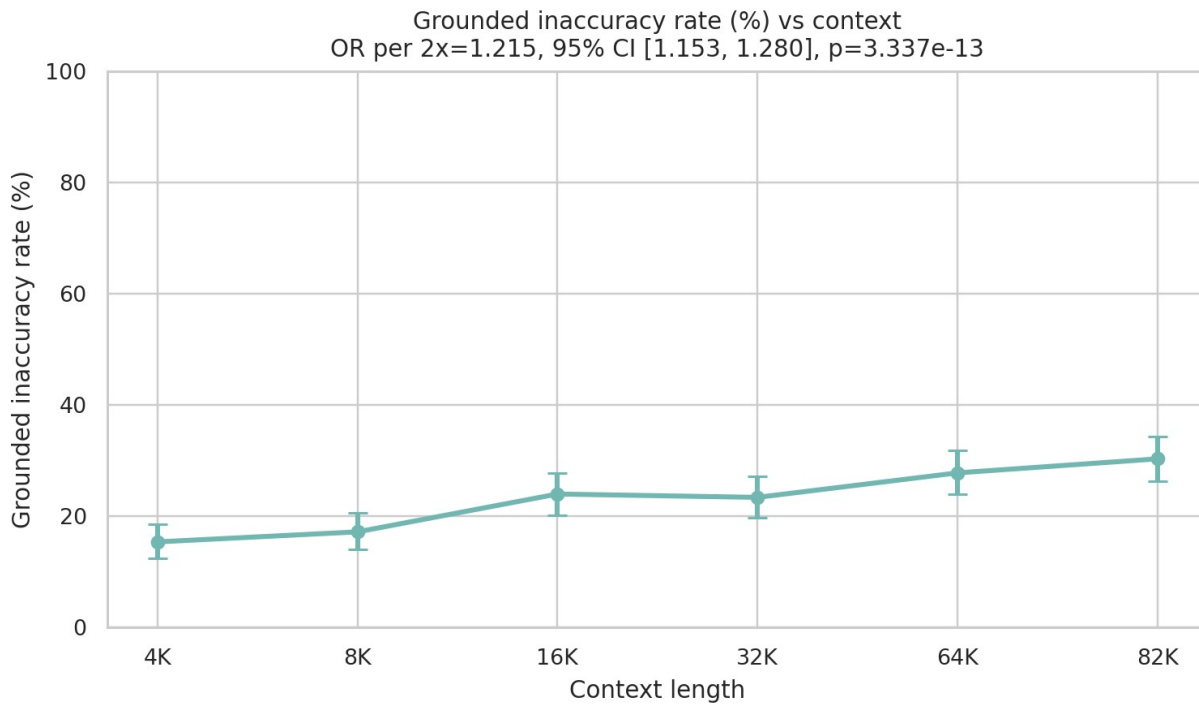


Figure 4. Grounded Inaccuracy vs Context. GEE OR = 1.215, 95% CI [1.153, 1.280], $p = 3.34 \times 10^{-13}$.

12. Sensitivity Analyses

12.1 Excluding UNANSWERABLE Questions

Table 7. Trend models after excluding UNANSWERABLE families.

Outcome	OR	95% CI	p-value
inaccurate	1.129	[1.081, 1.180]	5.30e-08
hallucinatory_inaccuracy	0.942	[0.906, 0.980]	0.00275
grounded_inaccuracy	1.234	[1.166, 1.306]	2.66e-13

After excluding UNANSWERABLE families, overall inaccuracy and grounded inaccuracy still increased, while Hallucinatory Inaccuracy decreased. The full-dataset increase in Hallucinatory Inaccuracy is therefore driven primarily by failed abstention on unanswerable tasks.

12.2 Complete-Case Sensitivity

Table 8. Complete-case trend models using 498 families and 2,988 observations.

Outcome	OR	95% CI	p-value
inaccurate	1.232	[1.178, 1.289]	8.14e-20
hallucinatory_inaccuracy	1.072	[1.028, 1.117]	0.00105
grounded_inaccuracy	1.213	[1.151, 1.278]	4.69e-13

Conclusions were unchanged.

13. Paired McNemar Tests

All higher contexts were significant versus 4K for overall inaccuracy after Holm correction. Hallucinatory Inaccuracy was significant for 4K vs 8K, 64K, and 82K. Grounded Inaccuracy was significant for 4K vs 16K, 32K, 64K, and 82K.

Table 9. Paired McNemar comparisons against 4K.

Outcome	Comparison	Paired N	Rate Difference (pp)	4K=0, Higher =1	4K=1, Higher =0	Raw p	Holm p	Holm Significant
inaccuracy	4K_vs_8K	500	4.8	41	17	0.00223	0.00223	True
inaccuracy	4K_vs_16K	500	9.2	66	20	6.67e-07	1.33e-06	True
inaccuracy	4K_vs_32K	500	12.0	88	28	2.09e-08	6.27e-08	True
inaccuracy	4K_vs_64K	500	20.0	124	24	1.89e-17	7.57e-17	True
inaccuracy	4K_vs_82K	498	21.3	129	23	4.33e-19	2.17e-18	True
hallucinatory_inaccuracy	4K_vs_8K	500	3.0	24	9	0.01353	0.04059	True
hallucinatory_inaccuracy	4K_vs_16K	500	0.6	35	32	0.80719	0.80719	False
hallucinatory_inaccuracy	4K_vs_32K	500	4.0	51	31	0.03524	0.07048	False
hallucinatory_inaccuracy	4K_vs_64K	500	7.6	75	37	0.00042	0.0021	True
hallucinatory_inaccuracy	4K_vs_82K	498	6.4	69	37	0.00244	0.00976	True
grounded_inaccuracy	4K_vs_8K	500	1.8	39	30	0.33556	0.33556	False
grounded_inaccuracy	4K_vs_16K	500	8.6	68	25	9.38e-06	2.81e-05	True
grounded_inaccuracy	4K_vs_32K	500	8.0	70	30	7.85e-05	0.00016	True
grounded_inaccuracy	4K_vs_64K	500	12.4	90	28	8.91e-09	3.57e-08	True
grounded_inaccuracy	4K_vs_82K	498	14.9	96	22	3.26e-12	1.63e-11	True

14. Question-Type Analysis

Table 10. Question-type outcome rates at 4K and 82K.

Question Type	4K Inaccuracy	82K Inaccuracy	4K Hallucinat ory	82K Hallucinat ory	4K Grounded	82K Grounded
DIRECT_RETRIEVAL	24.0%	37.0%	0.0%	1.0%	24.0%	36.0%
ENTITY_UNIT_BINDING	25.3%	50.5%	4.2%	2.1%	21.1%	48.4%
RETRIEVAL_CALCULATION	94.0%	89.9%	78.0%	64.9%	16.0%	25.0%
TEMPORAL_VERSION	18.2%	67.3%	1.8%	9.1%	16.4%	58.2%
UNANSWERABLE	49.0%	97.0%	49.0%	97.0%	0.0%	0.0%

UNANSWERABLE failures rose from approximately 49.0% to 97.0%, explaining much of the full-dataset Hallucinatory Inaccuracy increase. TEMPORAL_VERSION inaccuracy rose from approximately 18.2% to 67.3%, indicating temporal/version confusion. These subgroup results are exploratory.

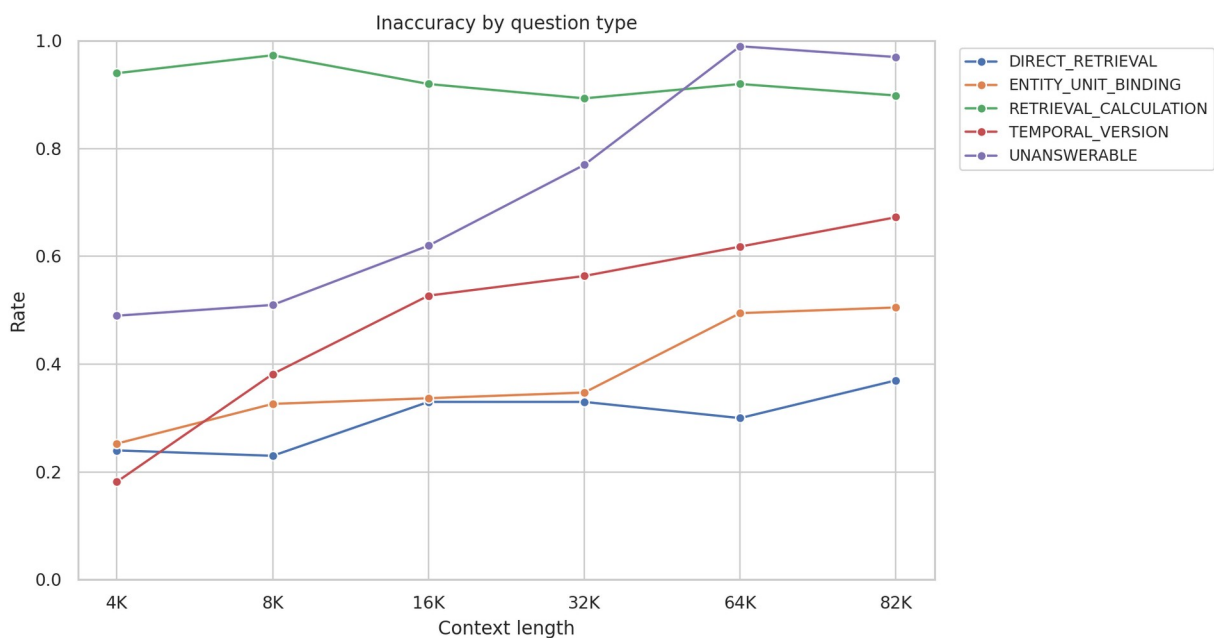


Figure 6. Inaccuracy by question type and context length.

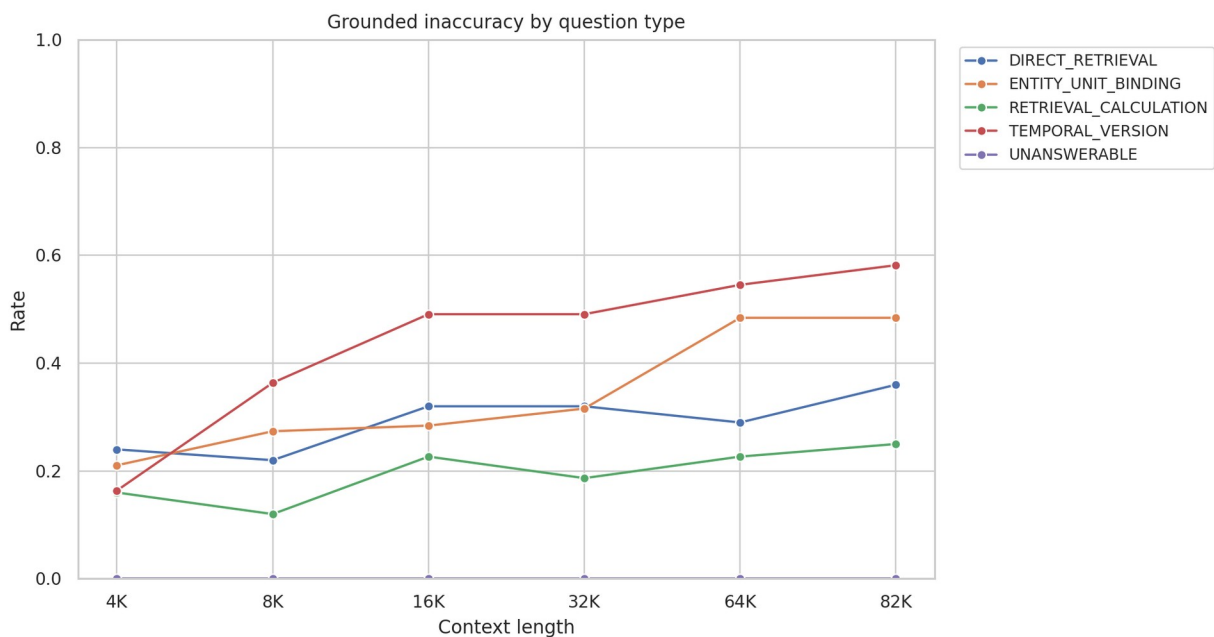


Figure 7. Grounded Inaccuracy by question type and context length.

15. Domain Analysis

Table 11. Domain outcome rates at 4K and 82K.

Domain	4K Inaccuracy	82K Inaccuracy	4K Hallucination	82K Hallucination	4K Grounded	82K Grounded
CLINICAL_TRIALS	44.0%	59.2%	40.8%	36.8%	3.2%	22.4%
FDA	58.4%	62.4%	53.6%	44.8%	4.8%	17.6%
FRED	43.2%	79.0%	20.8%	36.3%	22.4%	42.7%
SEC	52.8%	82.3%	21.6%	43.5%	31.2%	38.7%

SEC and FRED showed the largest overall inaccuracy increases. FDA was comparatively flatter. ClinicalTrials increased mainly through grounded errors. Domain-level results are exploratory.

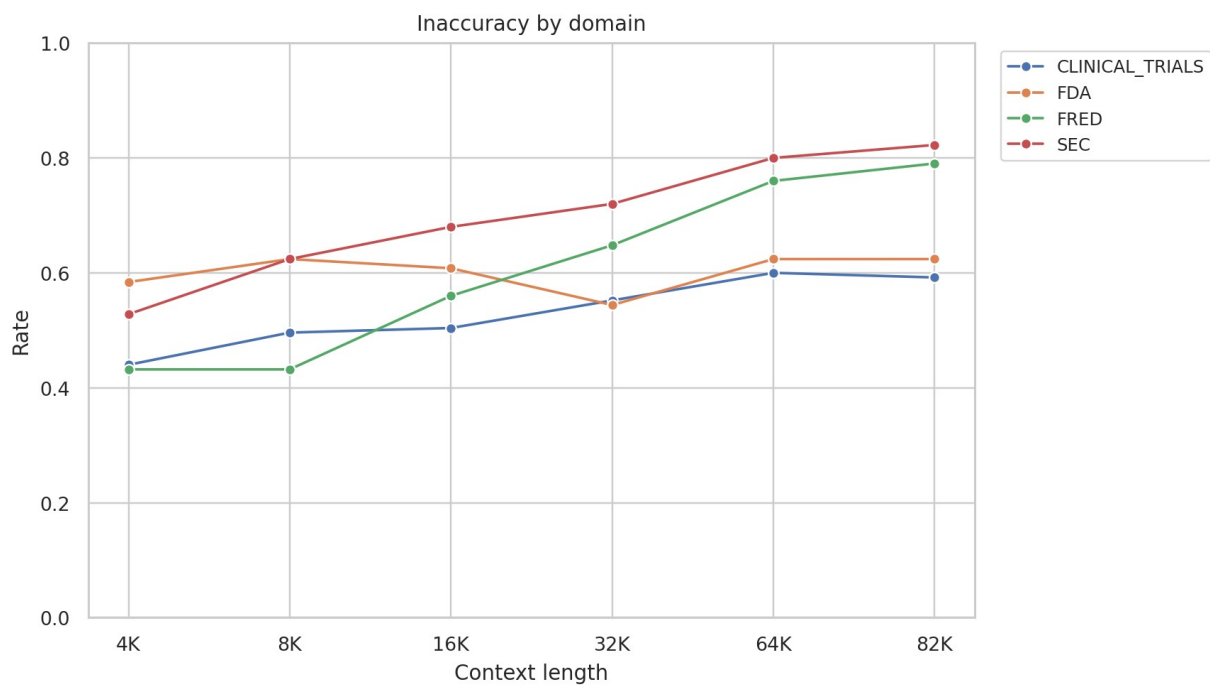


Figure 8. Inaccuracy by domain and context length.

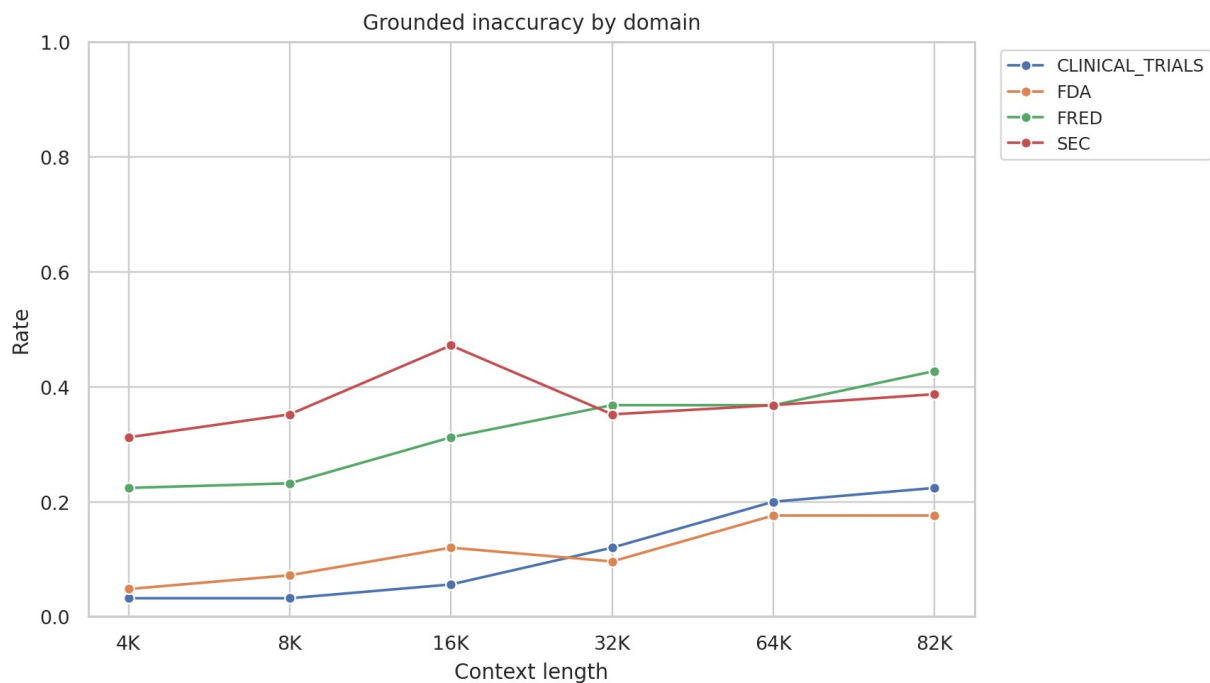


Figure 9. Grounded Inaccuracy by domain and context length.

16. Error-Type Evolution

Table 12. Detailed error-type counts by context.

Error Type	4K	8K	16K	32K	64K	82K	82K - 4K
CALCULATION_ERROR	12	9	8	12	14	13	1

FAILED_TO_ABSTAIN	49	51	62	77	99	97	48
UNNECESSARY_ABSTENTION	11	5	20	8	12	10	-1
UNSUPPORTED_VALUE	122	135	112	114	110	104	-18
WRONG_ENTITY	32	45	44	42	43	47	15
WRONG_FIELD	1	1	4	6	12	15	14
WRONG_PERIOD	6	10	19	19	35	45	39
WRONG_SERIES_VARIANT	2	3	2	3	1	2	0
WRONG_UNIT	3	2	0	2	0	1	-2
WRONG_VERSION	10	11	23	25	22	18	8

The largest 4K-to-82K increases were FAILED_TO_ABSTAIN (+48), WRONG_PERIOD (+39), WRONG_ENTITY (+15), and WRONG_FIELD (+14). UNSUPPORTED_VALUE decreased by 18. The growth in total inaccuracy is therefore substantially driven by abstention failure and contextual misbinding.

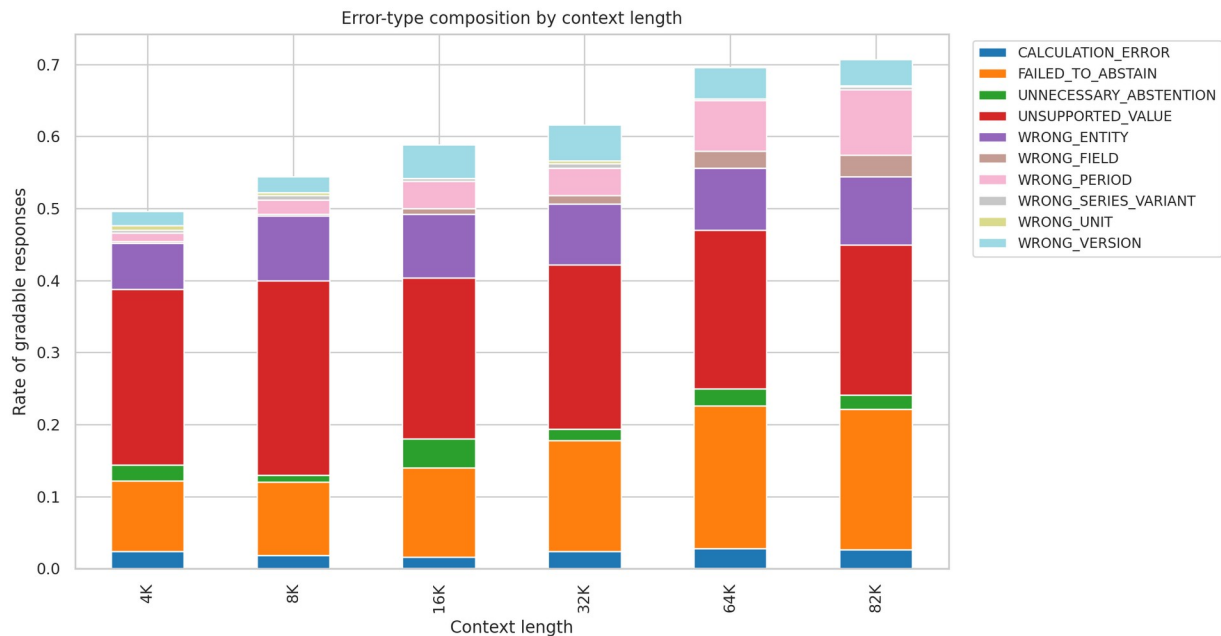


Figure 5. Error-type composition by context length.

17. Latency

Table 13. Inference latency by context.

Context	N	Mean	Median	P95	P99	Mean Input	Mean Output
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						Tokens	Tokens
4K	500	0.316 s	0.315 s	0.350 s	0.355 s	4,273.3	8.4
8K	500	0.598 s	0.597 s	0.635 s	0.641 s	8,330.5	8.3
16K	500	1.212 s	1.211 s	1.260 s	1.268 s	16,442.3	8.3
32K	500	2.842 s	2.832 s	2.938 s	2.944 s	32,671.4	8.3
64K	500	7.827 s	7.804 s	7.959 s	7.968 s	65,126.1	8.2
82K	498	11.260 s	11.251 s	11.385 s	11.409 s	81,745.1	8.3

Inference latency increased sharply with context length. This report does not claim latency causes factual errors.

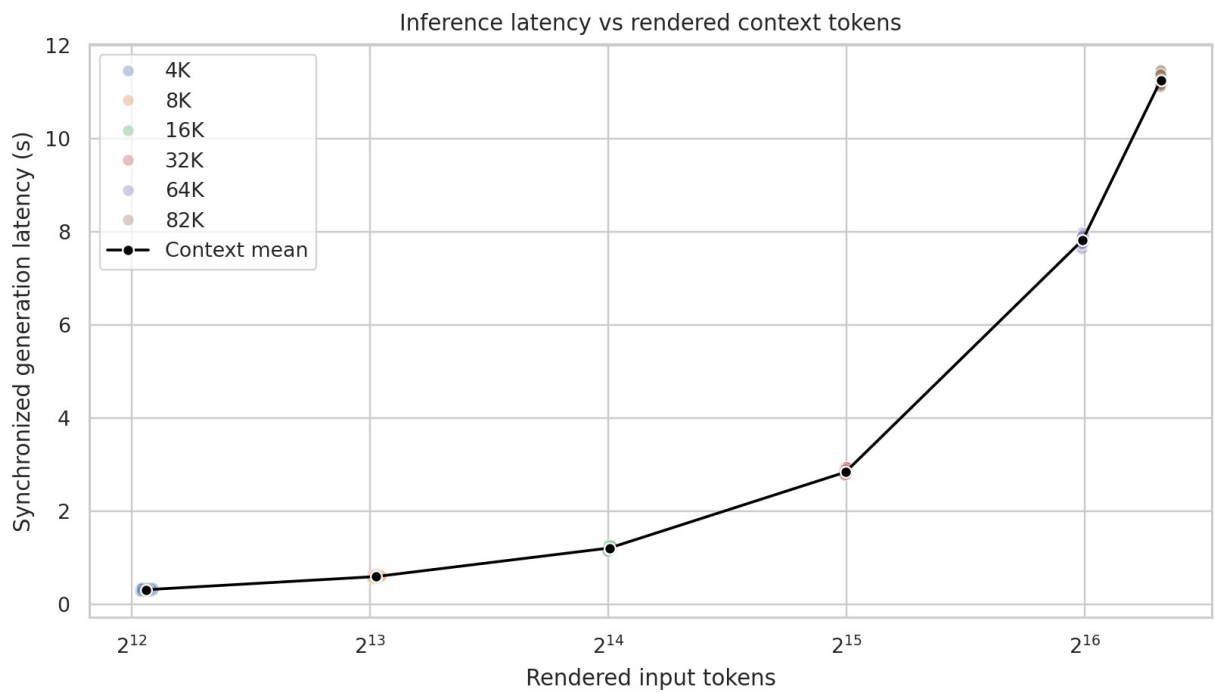


Figure 10. Inference latency vs rendered context tokens.

18. Family-Level Transitions

Family-level trajectories were heterogeneous, including always-correct, always-inaccurate, first-failure-at-longer-context, and non-monotonic patterns.

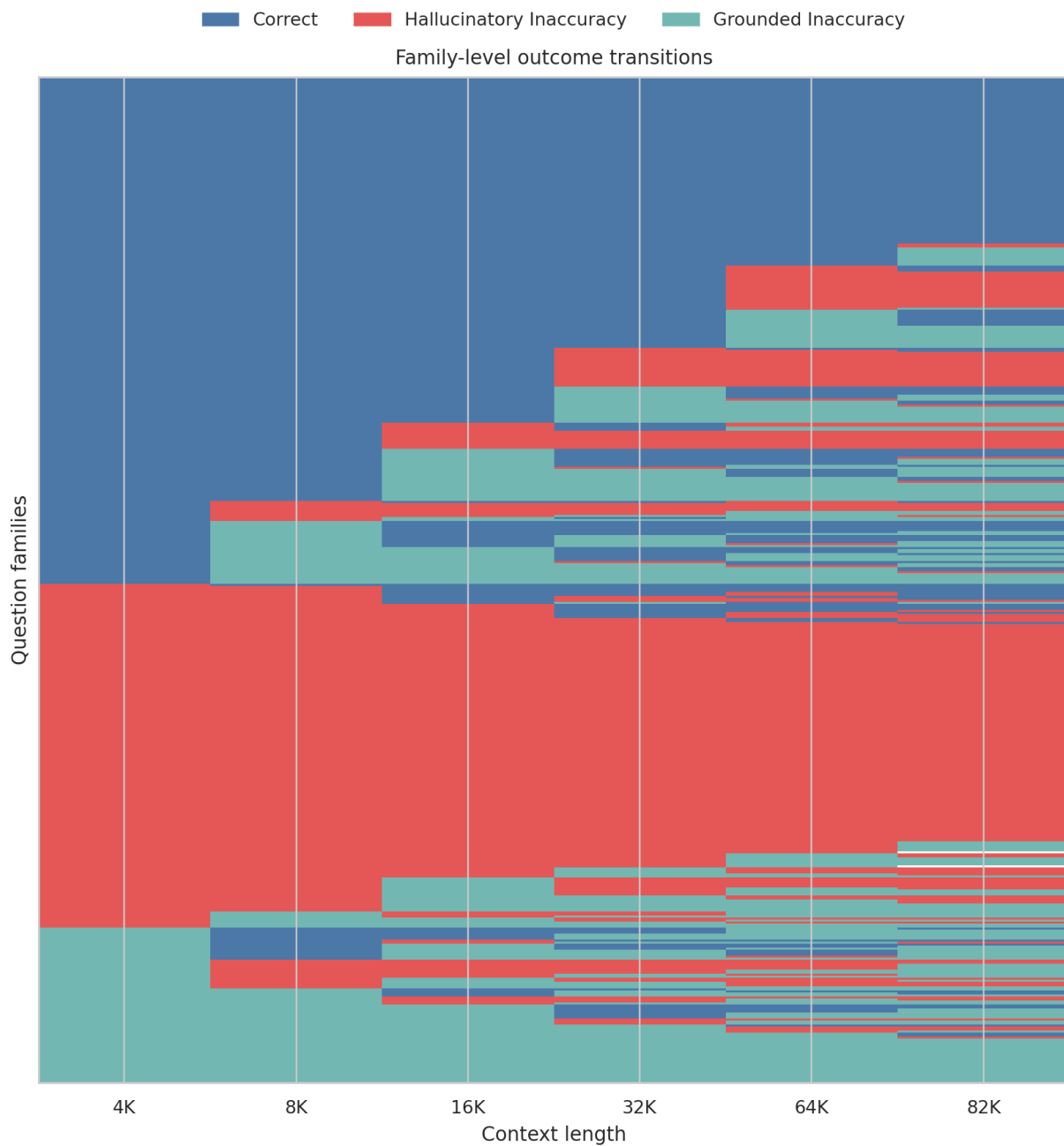


Figure 11. Family-level transition visualization across context lengths.

19. Discussion

Increasing context length substantially reduces factual reliability. The degradation is not captured adequately by a single hallucination metric. More context introduces more authentic competing information, and the model increasingly selects, binds, or reasons over legitimate but incorrect contextual records. This grounded contextual-confusion mechanism is distinct from classical unsupported hallucination.

The full dataset shows increasing Hallucinatory Inaccuracy, but the sensitivity analysis demonstrates that this is driven primarily by failed abstention on unanswerable questions. On answerable factual tasks, Hallucinatory Inaccuracy decreases while Grounded Inaccuracy increases.

20. Limitations

Limitations include one model, one hardware/inference configuration, 500 question families across four structured factual domains, context tested through approximately 82K rendered

input tokens, two 82K CUDA OOM runtime failures, answer-only output with no evidence-selection metric, deterministic grading plus 19 manually adjudicated edge cases, high-quality competing distractors by design, exploratory subgroup analyses, GEE rather than a full GLMM, and uncertain generalization to larger or proprietary LLMs.

21. Conclusion

Inaccuracy increased from 49.6% at 4K to 70.7% at 82K. The primary GEE model estimated an inaccuracy OR of 1.232 per context doubling, $p = 8.66 \times 10^{-20}$. Grounded Inaccuracy increased from 15.4% to 30.3%, OR 1.215, $p = 3.34 \times 10^{-13}$. Hallucinatory Inaccuracy increased from 34.2% to 40.4% in the full dataset, OR 1.070, $p = 0.00133$, but excluding unanswerable tasks reversed that trend, OR 0.942, $p = 0.00275$.

Increasing context length substantially reduces factual reliability. The most robust mechanism on answerable factual tasks is an increase in grounded contextual confusion rather than an increase in unsupported fabrication.

Appendix A. Reproducibility and Provenance

Final JSONL hash:

8fa4fd3b990adf54b6f7790ed6defa9c5d89aa6dd8365e0a7519beeb44d985e8

Final CSV hash: 155f80ec3bf284a7928ede0d24b491a972540b77dfe01d98eb622825c9c06a78

Benchmark hash:

dc2c4194dedb090198e6883735257908ce274bebc8611b40d958dbd026aa1fe6

Grader hash: d9282a0ccc50daba3bfd232c058dfbab63a5c19a7323d585a7c2236b3a6c4ba8

Prompt hash: 5d2869822989e19b

Model revision: 0cb88a4f764b7a12671c53f0838cd831a0843b95